# Supplementary material

*For "When Is a Call–Put Implied-Volatility Spread Measurable? An Abstention-Aware Protocol Validated Against a Known Truth". These appendices are not typeset with the article; they are published online alongside it.*

## Table S1. Certified coverage by strata, development draw

| Family | Bucket | n | Certified coverage (%) | Certified MAE (pp) | Raw-midpoint MAE (pp) | Raw MAE on abstained (pp) |
|---|---|---|---|---|---|---|
| Moneyness | K/S=0.90 | 179 | 0.0 | — | 0.054 | 0.054 |
| Moneyness | K/S=0.95 | 179 | 13.4 | 0.010 | 0.028 | 0.030 |
| Moneyness | K/S=1.00 | 179 | 13.4 | 0.009 | 0.024 | 0.027 |
| Moneyness | K/S=1.05 | 178 | 13.5 | 0.007 | 0.041 | 0.047 |
| Moneyness | K/S=1.10 | 179 | 0.0 | — | 2.508 | 2.508 |
| Maturity | 14d | 298 | 0.0 | — | 1.504 | 1.504 |
| Maturity | 30d | 298 | 0.0 | — | 0.068 | 0.068 |
| Maturity | 60d | 298 | 24.2 | 0.009 | 0.019 | 0.022 |
| Half-spread | 1 tick half-spread | 299 | 24.1 | 0.009 | 0.020 | 0.023 |
| Half-spread | 2 tick half-spread | 348 | 0.0 | — | 0.062 | 0.062 |
| Half-spread | 3 tick half-spread | 99 | 0.0 | — | 3.568 | 3.568 |
| Half-spread | wide session | 148 | 0.0 | — | 0.644 | 0.644 |
| True magnitude | 0.1-1 pp | 299 | 16.1 | 0.010 | 0.022 | 0.024 |
| True magnitude | <0.1 pp | 238 | 0.0 | — | 0.052 | 0.052 |
| True magnitude | >=1 pp | 357 | 6.7 | 0.007 | 1.275 | 1.366 |

*Note: n is the number of attempted pairs in the bucket. Coverage is the share of the bucket the protocol certifies at 0.10 percentage points. MAE is the mean absolute error against the benchmark's defined truth, in percentage points of volatility, computed on the pairs each method can use; a dash means the method certified nothing in that bucket. The rightmost column is the raw midpoint's error on the pairs the protocol declined to certify.*

## Table S2. The declared settings

| Setting | Value |
|---|---|
| Inversion map | European Black–Scholes–Merton, common to both legs |
| Root bracket | decimal IV in $[10^{-8}, 5]$, bisection, $\leq 64$ iterations, no clipping |
| Near-bound guard | prices within $10^{-10}$ of a European bound are rejected, not inverted |
| Resolution scale | 1 tick of out-of-the-money-equivalent time value |
| Finite shock | one tick of the quote grid (0.01 in the base market, 0.05 in the coarse-tick specification), applied to each leg's midpoint price, up and down |
| Synchronization tolerance | 2 s for call–put, call–spot and put–spot lags |

| Setting | Value |
|---|---|
| Tolerance grid | 0.05 / 0.10 / 0.25 percentage points, all reported |
| Primary tolerance | 0.10 percentage points, declared in the addendum |
| Certification displacement | the larger distance from the midpoint CPIV to either endpoint of the endpoint range is at most the tolerance |
| Sign band | 0.05 percentage points; the sign-disagreement count omits truths smaller in magnitude than the band |
| Perturbation set | endpoint widened by half a tick; spot +1 bp; rate +5 bp; spot timestamp −3 s; resolution scale halved to 0.5 tick; synchronization tolerance relaxed to 5 s |
| Period split | primary = days 0–14, holdout = days 15–29 |
| Inference unit | underlying-day block, 2 000 resamples, seed 20261022 |

*Note: every value is declared in the addendum, dated 22 October 2026, the same day as the sweep manifest; Section 5.1 gives the provenance.*

# Appendix A. The protocol and the instrument in one page

**A.1 Stages.** Contract match → timing check → quote validity → input validity → midpoint bounds → endpoint inversion → time-value resolution → finite-shock resolution → tolerance certification.

**A.2 Statuses.** Seven terminal statuses, mutually exclusive, summing to the attempted count. Reason codes stored separately and never substituted for a status.

**A.3 Inversion.** European Black–Scholes–Merton for both legs, decimal IV in $[10^{-8}, 5]$, bisection with at most 64 iterations, no clipping, near-bound guard at $10^{-10}$.

**A.4 Resolution.** Time value of at least the resolution scale; valid roots at both shocks of one tick applied to each leg's midpoint; a finite endpoint range over the four bid–ask combinations, one-sided where an endpoint has no root, never imputed.

**A.5 Certification.** A pair is certified when its displacement, the larger of the distances from the midpoint CPIV to the two endpoints of its endpoint range, is at most the declared tolerance, and the three preceding rules of Section 3.3 are also satisfied.

**A.6 Inference.** Underlying-day block bootstrap, 2 000 resamples, seed 20261022. The same block device is used for the null in the selection frontier.

**A.7 Generator.** Two underlyings at spot 100 and 250; 30 trading days each; maturities of 14, 30 and 60 days; strikes at 90 %, 95 %, 100 %, 105 % and 110 % of spot; a continuously compounded rate of 5 % and no dividend yield; base volatility of 22 % with a term slope of +0.12 percentage points per day and a log-moneyness skew of −1.10; a tick of 0.01, a half-spread of one tick, one extra tick in the wings and at the short end, doubled on every seventh session; timestamp jitter of ±0.35 s; and injected defects at declared rates — stale quotes 1.5 %, crossed quotes 0.8 %, out-of-bounds quotes 0.5 %, contract mismatches 0.5 % and missing legs 0.4 %. The put leg is priced on an 800-step binomial lattice, so the truth carries a genuine early-exercise premium.

# Appendix B. The replication design in one page

**B.1 Seeds.** 23 replication seeds, 20261022 + 7919k for k = 1 … 23, plus the development draw at k = 0, excluded from every tally except R3 and R6, which are computed on its seed.

**B.2 Specifications.** 17, one factor at a time from the base market: nine market specifications, seven protocol specifications and the base itself.

**B.3 Frontier.** 2 000 Monte Carlo trials at each of seven evaluation points — 0.05, 0.075, 0.10, 0.15, 0.20, 0.25 and 0.50 percentage points — against a pair null (uniform without replacement) and a day-block null (whole underlying-days with replacement, partial final block taken at random). Add-one p-values. The tolerance grid of Table S2 is 0.05, 0.10 and 0.25 percentage points. The points 0.075, 0.15 and 0.20 are added only so the curve can be drawn. The point 0.50 is also a tolerance specification in the sweep (Section 5.3). Certification and the reported coverages use the grid alone, and the decision rules of the addendum read the frontier at the primary tolerance only, 0.10 pp. Table 3 shows the grid points that certify plus two intermediate points; the other rows are in the released artifact.

**B.4 Tally.** `verdict.py` encodes the six declared rules and reads only the sweep and study outputs.

**B.5 Artifacts.** The run manifests record the configuration, the seed, the environment and the output hashes of every run reported in the article. The per-draw sweep outputs, the specification grid and the rule tally are released as aggregate CSV and JSON files; no market data is included because none was used.